Experiment No. 2

Title: E-Commerce System (Customer Account ID)

Objectives:
- To implement searching algorithms in python
- To store and manage customer account IDs in an E-Commerce System.
- To search for a specific customer account ID efficiently.

2. Problem Statement:

Develop a python program for an E-commerce System to store Customer account IDs and search for a particular customer account ID using searching algorithms.

3. Outcomes:
- A python program demonstrating searching algorithms.
- Successful storage and searching of customer account IDs.
- Understanding of searching techniques in real-world applications.

4. Software & Hardware requirements:
Software: Python 3.x Interpreter

Hardware: A computer capable of running python programs.

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5. Theory / concept in Brief:

Linear search:

This algorithm checks each element of the one by one until the required element is founder or the list ends.

Binary search:

This algorithm searches for an element in a sorted list by repeatedly dividing the search interval into two halves.

6. Algorithm:

1. Linear Search:

- Initialize the list of customer account IDs
- Accept the customer account ID to sear
- If a match is found, display the custom ID as found.

2. Binary search

- Arrange customer account IDs in ascending
- Accept the customer account ID to search
- Display whether the ID is found or not

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7. Test case

Case 1:

Input: Customer IDs = [101, 102, 103, 104, 105]

search ID = 103

Expected output: Customer Account ID found.

Case 2:

Input: Customer IDs = [101, 102, 103, 104, 105]

search ID = 110

Expected output: Customer Account ID not found.

8. Mathematical Model:

The searching process checks whether the given customer account ID exists in the list.

- Linear search: Best case O(1), Worst case O(n).
- Binary search: Best case O(1), Worst case O(log n).

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9. Conclusion / Analysis :

This practical demonstrates the application of Linear Search & Binary Search in an E-Commerce System for searching customer account IDs. Both algorithms help identify customer records. Binary search is more efficient for large, sorted datasets, whereas Linear search can be used on unsorted lists.